

Malware Analysis Report For elf-Linux-ARMv7-Is

Binary Type

ELF (Executable and Linkable Format). (Approved)

Analyzing File

- **File Name:** elf-Linux-ARMv7-Is
 - **File Extension:**
 - **File Type:** ELF (Executable and Linkable Format).
 - **MD5 Hash:** de9f91f9cd038989fec8abf25031b42b
 - **SHA256 Hash:** e68c62b49184ed764f324fb4722481d60e1bf321b722238d95247f391960605c
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File Identification

File detected as a valid Linux executable.

ELF File Analysis

ELF File Analysis: True

ELF File Imports

- libc.so.2
- libacl.so.1
- libgcc_s.so.1
- libc.so.6
- ld-linux-armhf.so.3

ELF File Exports

No Exported functions were found.

String Analysis

ASCII Strings

- /lib/ld-linux.so.3
- vblCw)
- libc.so.2
- /TMderegisterTMCloneTable
- gmon_start
- JvRegisterClasses
- /TMregisterTMCloneTable
- capgetfile
- cap_free
- captotext

- libacl.so.1
- *aclgetentry*
- *aclgettag_type*
- *aclextendedfile*
- libgcc_s.so.1
- *_aeabiunwindcpppr0*
- libc.so.6
- fflush
- strcpy
- *_printfchk*
- fnmatch
- setlocale
- mbrtowc
- strncmp
- optind
- strchr
- fflush_unlocked
- dcgettext
- stpncpy
- getpwuid
- closedir
- *_mempcpychk*
- getgrgid
- signal
- mbstowcs
- sigprocmask
- *_stackchk_fail*
- iswprint
- realloc
- *programinvocationname*
- strftime
- *_assertfail*
- *_ctypegetmbcur_max*
- isatty
- getpwnam
- calloc
- strlen
- ungetc
- sigemptyset
- memset
- localeconv
- strstr
- *_ermolocation*
- open64
- memcmp
- memcpy
- *__fxstat64*
- _setjmp
- *_fprintfchk*
- sigaddset
- getgrnam

- wcswidth
- stdout
- fseeko64
- memcpy
- fclose
- strtoul
- malloc
- mbsinit
- __lxstat64
- __uflow
- nl_langinfo
- opendir
- __xstat64
- _ctypeb_loc
- getenv
- optarg
- __freanding
- stderr
- wcwidth
- *obstack*newchunk
- readlink
- fscanf
- getopt_long
- fileno
- fwrite
- gettimeofday
- sigaction
- *_memcpychk*
- sigismember
- clock_gettime
- __fpending
- localtime
- lseek64
- strchr
- iswcntrl
- mktime
- program*invocations*short_name
- fdopen
- readdir64
- wcstombs
- *_ctype*toupper_loc
- *_ctype*tolower_loc
- *_sprintfchk*
- memmove
- *obstack*begin
- bindtextdomain
- fwrite_unlocked
- strcmp
- *_strtol*internal
- tcgetpgrp
- *_libc*start_main
- stpcpy

- `strcoll`
- `__overflow`
- `fputs_unlocked`
- `__progname`
- `__prognamefull`
- `_cxa_atexit`
- `ld-linux-armhf.so.3`
- `__stackchk_guard`
- `_edata`
- `__bssstart`
- **`bss_start`**
- **`bss_end`**
- **`end`**
- `ACL_1.0`
- `GLIBC_2.4`
- `GLIBC_2.17`
- `GCC_3.5`
- `sort_files`
- `format_inode`
- `?pcdb-lswd`
- `posix-`
- `devinopop`
- `sorttype != sortversion`
- `src/lis.c`
- $(((((\text{sizeof}(\text{uintmax_t}) * 8 - (!((\text{typeof}(\text{uintmax_t})) 0 < (\text{typeof}(\text{uintmax_t})) - 1))) * 146$
- `%s %s`
- `%s, %s`
- cannot access %s
- cannot read symbolic link %s
- Try '%s --help' for more information.
- Usage: %s [OPTION]... [FILE]...
- List information about the FILES (the current directory by default).
- Sort entries alphabetically if none of `-ctuvSUX` nor `--sort` is specified.
- Mandatory arguments to long options are mandatory for short options too.
- `-a`, `--all` do not ignore entries starting with `.`
- `-A`, `--almost-all` do not list implied `.` and `..`
- `--author` with `-l`, print the author of each file
- `-b`, `--escape` print C-style escapes for nongraphic characters
- `--block-size=SIZE` scale sizes by SIZE before printing them. E.g.,
- `'--block-size=M'` prints sizes in units of
- 1,048,576 bytes. See SIZE format below.
- `-B`, `--ignore-backups` do not list implied entries ending with `~`
- `-c` with `-lt`: sort by, and show, ctime (time of last
- modification of file status information)
- with `-l`: show ctime and sort by name
- otherwise: sort by ctime, newest first
- `-C` list entries by columns
- `--color[=WHEN]` colorize the output. WHEN defaults to 'always'
- or can be 'never' or 'auto'. More info below
- `-d`, `--directory` list directory entries instead of contents,
- and do not dereference symbolic links
- `-D`, `--dired` generate output designed for Emacs' dired mode

- -f do not sort, enable -aU, disable -ls --color
- -F, --classify append indicator (one of */=>@|) to entries
- --file-type likewise, except do not append '**'
- --format=WORD across -x, commas -m, horizontal -x, long -l, single-column -1, verbose -l, vertical -C
- --full-time like -l --time-style=full-iso
- -g like -l, but do not list owner
- --group-directories-first
- group directories before files.
- augment with a --sort option, but any
- use of --sort=none (-U) disables grouping
- -G, --no-group in a long listing, don't print group names
- -h, --human-readable with -l, print sizes in human readable format (e.g., 1K 234M 2G)
- --si likewise, but use powers of 1000 not 1024
- -H, --dereference-command-line
- follow symbolic links listed on the command line
- --dereference-command-line-symlink-to-dir
- follow each command line symbolic link
- that points to a directory
- --hide=PATTERN do not list implied entries matching shell PATTERN (overridden by -a or -A)
- --indicator-style=WORD append indicator with style WORD to entry names:
- none (default), slash (-p),
- file-type (--file-type), classify (-F)
- -i, --inode print the index number of each file
- -I, --ignore=PATTERN do not list implied entries matching shell PATTERN
- -k, --kibibytes use 1024-byte blocks
- -l use a long listing format
- -L, --dereference when showing file information for a symbolic link, show information for the file the link
- references rather than for the link itself
- -m fill width with a comma separated list of entries
- -n, --numeric-uid-gid like -l, but list numeric user and group IDs
- -N, --literal print raw entry names (don't treat e.g. control characters specially)
- -o like -l, but do not list group information
- -p, --indicator-style=slash
- append / indicator to directories
- -q, --hide-control-chars print ? instead of non graphic characters
- --show-control-chars show non graphic characters as-is (default unless program is 'ls' and output is a terminal)
- -Q, --quote-name enclose entry names in double quotes
- --quoting-style=WORD use quoting style WORD for entry names: literal, locale, shell, shell-always, c, escape
- -r, --reverse reverse order while sorting
- -R, --recursive list subdirectories recursively
- -s, --size print the allocated size of each file, in blocks
- -S sort by file size
- --sort=WORD sort by WORD instead of name: none -U,
- extension -X, size -S, time -t, version -v
- --time=WORD with -l, show time as WORD instead of modification

- time: atime -u, access -u, use -u, ctime -c,
- or status -c; use specified time as sort key
- if --sort=time
- --time-style=STYLE with -l, show times using style STYLE:
- full-iso, long-iso, iso, locale, +FORMAT.
- FORMAT is interpreted like 'date'; if FORMAT is
- FORMAT1FORMAT2, FORMAT1 applies to
- non-recent files and FORMAT2 to recent files;
- if STYLE is prefixed with 'posix-', STYLE
- takes effect only outside the POSIX locale
- -t sort by modification time, newest first
- -T, --tabsize=COLS assume tab stops at each COLS instead of 8
- -u with -lt: sort by, and show, access time
- with -l: show access time and sort by name
- otherwise: sort by access time
- -U do not sort; list entries in directory order
- -v natural sort of (version) numbers within text
- -w, --width=COLS assume screen width instead of current value
- -x list entries by lines instead of by columns
- -X sort alphabetically by entry extension
- -Z, --context print any SELinux security context of each file
- -1 list one file per line
- --help display this help and exit
- --version output version information and exit
- SIZE is an integer and optional unit (example: 10M is 10⁷241024). Units
- are K, M, G, T, P, E, Z, Y (powers of 1024) or KB, MB, ... (powers of 1000).
- Using color to distinguish file types is disabled both by default and
- with --color=never. With --color=auto, ls emits color codes only when
- standard output is connected to a terminal. The LS_COLORS environment
- variable can change the settings. Use the dircolors command to set it.
- Exit status:
- 0 if OK,
- 1 if minor problems (e.g., cannot access subdirectory),
- 2 if serious trouble (e.g., cannot access command-line argument).
- Report %s bugs to %s
- bug-coreutils@gnu.org
- %s home page: <%s>
- GNU coreutils
- <http://www.gnu.org/software/coreutils/>
- General help using GNU software: <http://www.gnu.org/gethelp/>
- Report %s translation bugs to <http://translationproject.org/team/>
- For complete documentation, run: info coreutils '%s invocation'
- full-iso
- locale
- /usr/share/locale
- QUOTING_STYLE
- ignoring invalid value of environment variable QUOTING_STYLE: %s
- COLUMNS
- ignoring invalid width in environment variable COLUMNS: %s
- TABSIZE
- ignoring invalid tab size in environment variable TABSIZE: %s
- abcdghiklmnopqrstuwx:ABCDEFGHILNQRT:UXZ1

- invalid line width: %s
- invalid tab size: %s
- --sort
- --time
- --format
- --color
- --indicator-style
- --quoting-style
- Richard M. Stallman
- David MacKenzie
- LSBLOCKSIZE
- BLOCK_SIZE
- TIME_STYLE
- invalid time style format %s
- time style
- Valid arguments are:
- - [posix-]%s
- - +FORMAT (e.g., +%H:%M) for a 'date'-style format
- %Y-%m-%d %H:%M:%S.%N %z
- %Y-%m-%d %H:%M
- %Y-%m-%d
- %m-%d %H:%M
- error initializing month strings
- LSCOLORS
- *target*
- *sizeof (struct devino) <= extension* ({ struct obstack const *o = (&devinoobstack); (unsigne
- cannot open directory %s
- cannot determine device and inode of %s
- %s: not listing already-listed directory
- reading directory %s
- closing directory %s
- //DIREDD//
- //SUBDIREDD//
- //DIREDD-OPTIONS// --quoting-style=%s
- *hashgetnentries (activedirset) == 0*
- *unrecognized prefix: %s*
- *unparsable value for LSCOLORS* environment variable
- escape
- directory
- full-time
- group-directories-first
- human-readable
- kibibytes
- numeric-uid-gid
- no-group
- hide-control-chars
- reverse
- almost-all
- ignore-backups

- classify
- file-type
- dereference-command-line
- dereference-command-line-symlink-to-dir
- ignore
- indicator-style
- dereference
- literal
- quote-name
- quoting-style
- recursive
- format
- show-control-chars
- tabsize
- time-style
- block-size
- context
- author
- version
- extension
- access
- status
- verbose
- commas
- horizontal
- across
- vertical
- single-column
- always
- if-tty
- %b %e %Y
- %b %e %H:%M
- long-iso
- invalid argument %s for %s
- ambiguous argument %s for %s
- Valid arguments are:
- - %s
- write error
- %s: %s
- # entries: %lu
- # buckets: %lu
- # buckets used: %lu (%.2f%%)
- max bucket length: %lu
- KMGTPPEZY
- BLOCKSIZE
- POSIXLYCORRECT
- eEgGkKmMpPtTyYzZ0
- A NULL argv[0] was passed through an exec system call.
- /.libs/
- shell-always
- c-maybe

- `clocale`
- `%m/%d/%y`
- `%Y-%m-%d`
- `%H:%M:%S`
- `%s (%s) %s`
- License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>.
- This is free software: you are free to change and redistribute it.
- There is NO WARRANTY, to the extent permitted by law.
- Written by %s.
- Written by %s and %s.
- Written by %s, %s, and %s.
- Written by %s, %s, %s,
- and %s.
- Written by %s, %s, %s,
- %s, and %s.
- Written by %s, %s, %s,
- %s, %s, and %s.
- Written by %s, %s, %s,
- %s, %s, %s, and %s.
- Written by %s, %s, %s,
- %s, %s, %s, %s,
- and %s.
- Written by %s, %s, %s,
- %s, %s, %s, %s,
- %s, and %s.
- Written by %s, %s, %s,
- %s, %s, %s, %s,
- %s, %s, and others.
- Report bugs to: %s
- Copyright %s %d Free Software Foundation, Inc.
- memory exhausted
- `xstrtoul`
- `0 <= strtolbase && strtolbase <= 36`
- `lib/xstrtoul.c`
- invalid %s%s argument '%s'
- invalid suffix in %s%s argument '%s'
- %s%s argument '%s' too large
- `xstrtoumax`
- `/usr/lib`
- `CHARSETALIASDIR`
- `charset.alias`
- `%50s %50s`
- GCC: (GNU) 4.7.2
- GCC: (crosstool-NG 1.17.0) 4.7.2
- `.shstrtab`
- `.interp`
- `.note.ABI-tag`
- `.note.gnu.build-id`
- `.gnu.hash`
- `.dynsym`
- `.dynstr`
- `.gnu.version`

- `.gnu.versionr`
- `.rel.dyn`
- `.rel.plt`
- `.rodata`
- `.ARM.exidx`
- `.ehframe`
- `.initarray`
- `.fini_array`
- `.dynamic`
- `.comment`
- `.ARM.attributes`

UTF-8 Strings

No detected.

UTF-16LE Strings

No detected.

Extracted URLs

- <http://www.gnu.org/software/coreutils/>
- <http://www.gnu.org/gethelp/>
- <http://translationproject.org/team/>
- <http://gnu.org/licenses/gpl.html>

Base64 Strings

No valid Base64-encoded strings found.

File Entropy Analysis

- **Entropy:** 6.20
- **Status:** This file is likely packed or encrypted.

VirusTotal Analysis

Scan Date: 2024-10-22 11:54:30

Scan Results

- Malicious: 2
- Suspicious: 0
- Undetected: 63
- Harmless: 0
- Timeout: 0
- Confirmed-timeout: 0
- Failure: 0
- Type-unsupported: 12

Community Votes

- Harmless: 0
- Malicious: 0

Detailed Engine Results

Engine Name	Detection Category	Detection
Ikarus	malicious	Trojan.Linux.Agent
